ODT Summative 2

Project Design Document

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Section : C

1. Design Intent

- We are building a simplified robotic version of Rocky (from Project Hail Mary) along with a game app.
- It's a 5-limb spider-like creature that communicates through gestures and moves using its limbs.
- The app is like a maze game for Dr. Grace (from Project Hail Mary) to find Rocky.
- The main target audience is just robotics and movie and game enthusiasts, or anyone in general who enjoys interacting or playing with robots.

2. Inspiration & References

Idea Inspiration

- Project Hail Mary the movie, primary inspiration for Rocky's five-limbed body and gestural communication style.
- Rocky in the book moves using its back limbs for locomotion while the front limbs are used for interaction and expression.
- In the movie, there is a scene where Dr Grace needs to find Rocky. Our maze game is inspired by that scene.
- We mainly gained a basic understanding of how Rocky functions, its key features, and the movements we can showcase.

3. User Journey

Step-by-step experience of the user:

1. User starts with pressing Start to play the maze game.
2. As the user presses start, Rocky gives a thumbs up for good luck.
3. The up, down, left, and right buttons can be pressed by the user to move the cursor to reach Rocky.
4. If the user reaches Rocky on time, Robotic Rocky will give them a dab.
5. Else, Rocky gives them a fist bump for the effort.
6. And the game can be reset and then the player can play again.

4. Success Criteria

Define measurable success:

Function:

- The first 2 limbs are operating efficiently in terms of all the needed gestures.
- Ideal body balance is achieved.
- System responds according to the win/loss of the user
- Sound system works as planned.

User experience (easy to use, intuitive)

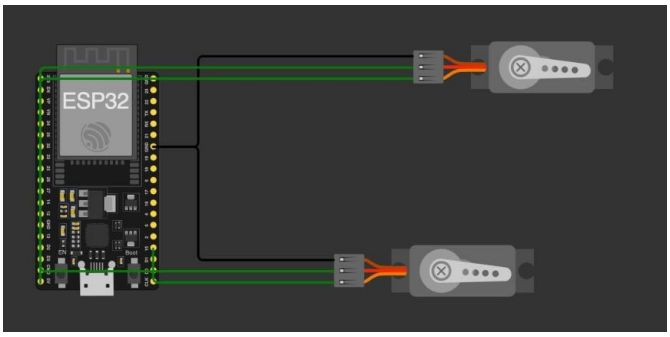
- The app UI is easy to navigate and operate.
- It is visually appealing.
- Should be fun to interact with.

5. System Overview

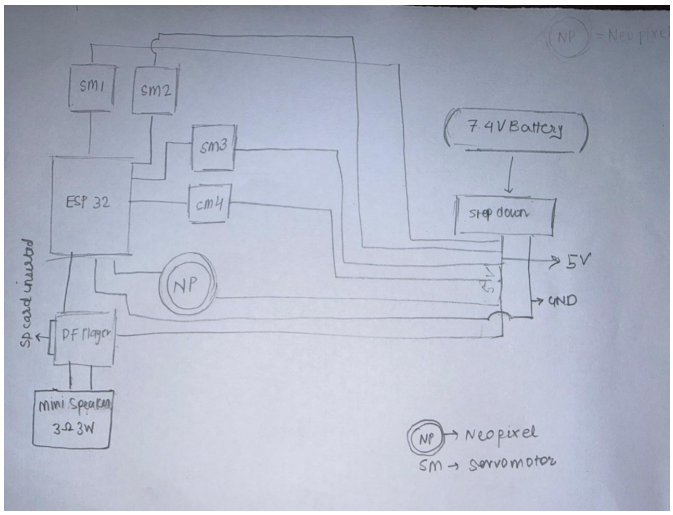
- We are making a robotic version of Rocky from the movie Project Hail Mary as well as a maze game for user experience.
- In our project, Rocky has 5 limbs – out of which only 2 will have movement for the hand gestures.
- We will be using 4 servo motors in total - 2 for each limb (elbows & thumbs).
- When the user presses the start button on the app, the robotic Rocky will give (and say) a thumbs up for good luck.
- The cursor will move according to the options the player chooses (up, down, left and right) and if the user's cursor reaches Rocky on time (40-60 seconds), the robotic Rocky will give a dab and say "Amaze, amaze, amaze!".
- However, if the player cannot reach Rocky on time, its a loss and the robotic Rocky will give a consolation fist bump and say "Fist my bump".
- As the player presses the reset button, the cursor goes back to the starting point and the player can click start to restart the game.

6. High-Level Block Diagram

Servo motors for the front limbs



Overall Diagram

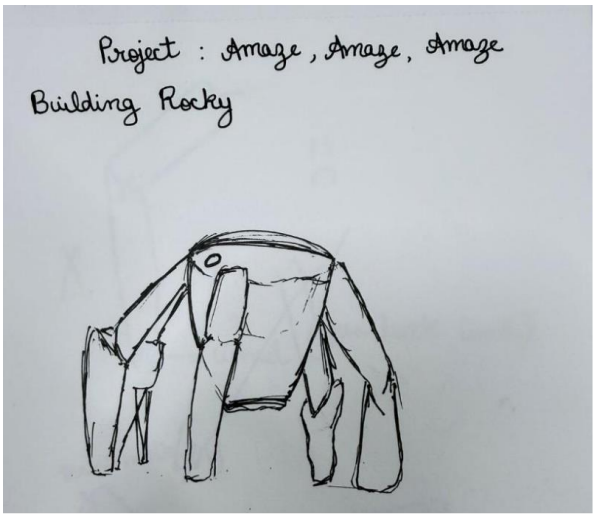


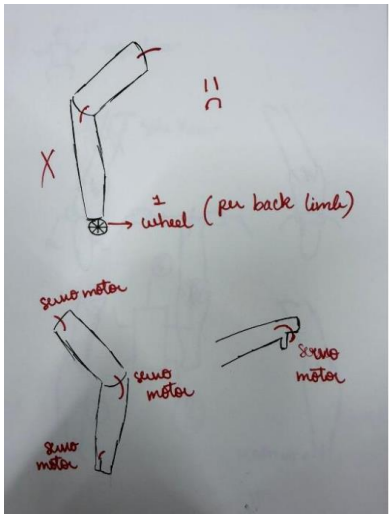
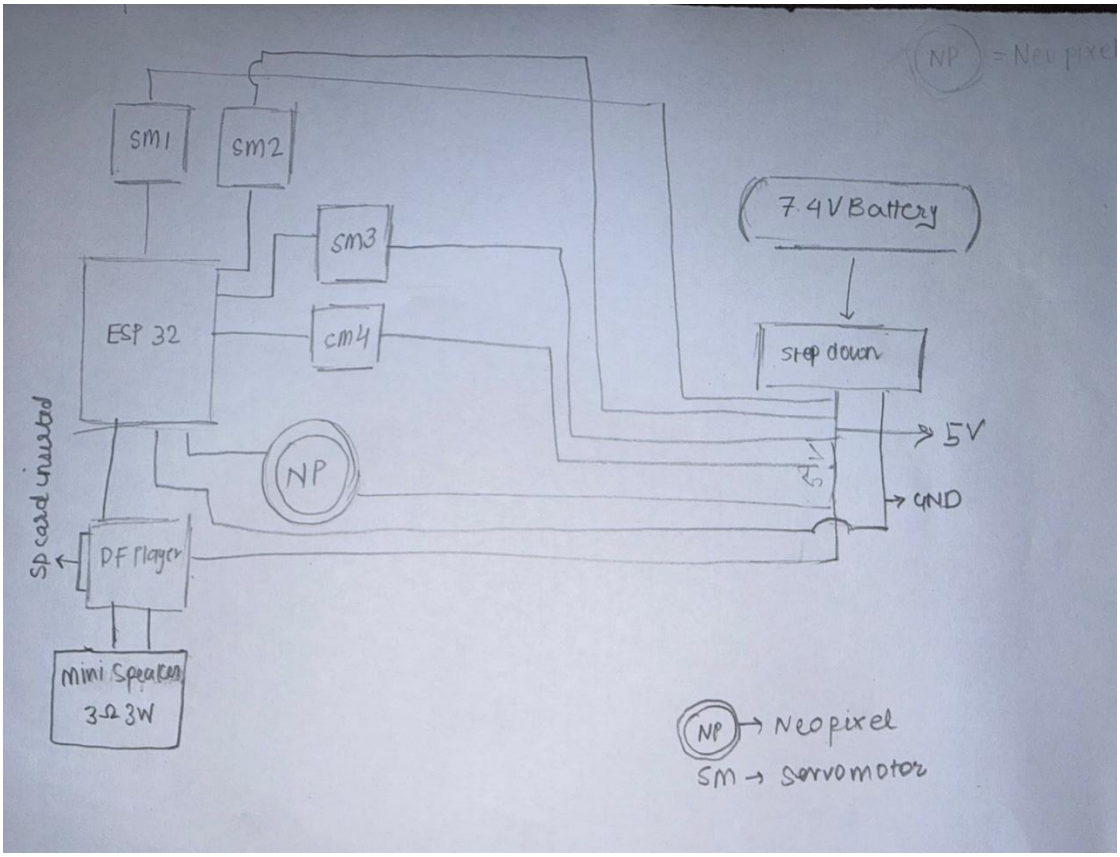
7. Input / Output Mapping

Input (Maze game app)	Process	Output
Scan button	The app code starts searching for nearby Bluetooth devices.	The app shows all the available Bluetooth devices.
Connect button	The app starts connecting to chosen device.	The device gets connected to the app.
Start button	The app code understands and processes the logic and the system receives the message.	The game begins, robotic Rocky says and gives a thumbs up and the timer starts.
Up, down, left & right buttons	The app code understands and processes the logic.	The cursor moves accordingly.
Cursor reaches Rocky	System receives the message	Robotic Rocky says “Amaze, amaze,

		amaze” and gives a dab to the user.
Cursor does not reach Rocky	System receives the message.	Robotic Rocky says “Fist my bump” and gives a consolation fist bump to the user.

8. Sketches & Concepts





9. Final System Architecture (Labelled)